

Robot Movement Based on IC3 Input Combinations

Input 1 (pin 2) of IC3	Input 2 (pin 7) of IC3	Input 3 (pin 10) of IC3	Input 4 (pin 15) of IC3	Movement of robot
Low	Low	Low	Low	No movement (M1-off, M2-off)
Low	High	Low	Low	Moves rightward (M1-runs, M2-off)
Low	Low	Low	High	Moves leftward (M1-off, M2-runs)
Low	High	Low	High	Moves straight (M1 and M2 runs)

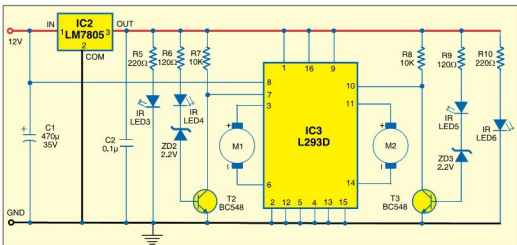


Fig. 3: Circuit diagram of path robot

10) are controlled through IR sensors. Based on the output of sensor IR LED4 and sensor IR LED5, low and high voltages can be provided to input pins 7 and 10 of IC3.

As per the input combinations shown in the table, the robot will move straight, left or right. When the robot approaches a wall, the IR light transmitted by IR LED3 or IR LED6 reflects off the wall to reach the receiving LED (IR LED4 or IR LED5). This generates a low output at pin 7 or pin 10 of IC3 to stop the respective motor (M1 or M2). And when there is no reflected light, i.e., no wall, the receiving LED (IR LED4 or IR LED5) makes the output of IC3 (at pin 7 or pin 10) high to start the motor (M1 or M2).

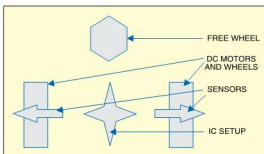


Fig. 4: Proposed assembly of robot

LED2 (refer Fig. 2).

Fig. 3 shows the circuit of the proximity sensor for path-tracking robots. The circuit makes use of proximity sensors IR LED3 through IR LED6, voltage regulator LM7805 (IC2), motor-driver

IC L293D (IC3), two geared motors/DC motors, 12V DC adaptor and two wheels. Voltage regulator IC2 converts the 12V DC into 5V DC. Pins 1 and 9 of IC3 are always kept high to enable the chip. Whenever inputs pins 2, 7, 10 and 15 of IC3 go high, the 12V DC (available at pin 8 of L293D) is transferred to the corresponding output pins (pins 3, 6, 11 and 14, respectively).

For controlling the path-tracking robot, input 1 (pin 2) and input 4 (pin 15) of IC3 are permanently held low, while input 2 (pin 7) and input 3 (pin

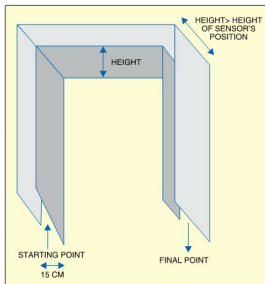


Fig. 5: Way of path tracker

VR1.

Working of the circuit is simple. When there is no object in front of IR LED1 and IR LED2, output pin 1 of IC1 remains low. As a result, LED1 remains off. When any object comes in front of IR LED1 and IR LED2, output pin 1 of IC1 goes high to light up LED1. At the same time, the relay-driver circuit (not shown here) is also activated.

Assemble the circuit on a general-purpose PCB and enclose in a small cabinet. Mount IR LED1 and IR LED2 such that when any obstacle comes in front of them, the IR reflected off the obstacle falls directly on receiver IR

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. After assembling the components, fix the free wheel on the front side and the two geared/DC motors on the opposite sides as shown in Fig. 4. Fig. 5 shows the working of the path tracker. ●